

Data Dictionary

NOTATION GUIDE

Symbol	Meaning
PK	Primary Key
FK	Foreign Key
UK	Unique Key
IDX	Indexed Field
REQ	Required (NOT NULL)
AUTO	Auto-generated
IMMUT	Immutable (cannot be updated after insert)
ENC	Encrypted field
AUDIT	Field tracked in audit log

1. USER MANAGEMENT & SECURITY

1.1 Table: users

Purpose: Store user accounts and authentication details

Field Name	Data Type	Size	Constraint s	Description	Example
id	BIGINT	-	PK, AUTO, REQ	Unique user identifier	1
employee_id	VARCHA R	20	UK, IDX	Employee ID from HR system	EMP-2024-001
username	VARCHA R	100	UK, REQ, IDX	Active Directory username	john.doe
email	VARCHA R	255	UK, REQ, IDX	Email address for notifications	john.doe@agency.g ov.ng
first_name	VARCHA R	100	REQ	User's first name	John
last_name	VARCHA R	100	REQ	User's last name	Doe
middle_name	VARCHA R	100	NULL	User's middle name	Michael
phone_number	VARCHA R	20	NULL	Mobile number for MFA	+234-803-123-4567

department_id	BIGINT	-	FK, REQ, IDX	Reference to departments table	5
zone_id	BIGINT	-	FK, REQ, IDX	Reference to zones table	2
is_active	BOOLEAN	-	REQ, DEFAULT true	Account status	true
is_locked	BOOLEAN	-	REQ, DEFAULT false	Account locked after failed attempts	false
failed_login_attempts	INT	-	REQ, DEFAULT 0	Count of consecutive failed logins	0
last_login_at	TIMESTAMP	-	NULL	Last successful login timestamp	2025-01-15 09:30:00
last_login_ip	VARCHAR	45	NULL	IP address of last login	192.168.1.100
password_changed_at	TIMESTAMP	-	NULL	Last password change date	2024-10-15 14:20:00

mfa_enabled	BOOLEAN	-	REQ, DEFAULT true	Multi-factor authentication enabled	true
mfa_secret	VARCHAR	ENC , 255	NULL, ENC	MFA secret key (TOTP)	[encrypted]
email_verified_at	TIMESTAMP	-	NULL	Email verification timestamp	2024-09-01 10:00:00
remember_token	VARCHAR	100	NULL	Remember me token	[token]
created_at	TIMESTAMP	-	REQ, AUTO	Record creation timestamp	2024-09-01 08:00:00
updated_at	TIMESTAMP	-	REQ, AUTO	Record last update timestamp	2025-01-15 09:30:00
created_by	BIGINT	-	FK, NULL	User who created this record	1
updated_by	BIGINT	-	FK, NULL	User who last updated this record	1
deleted_at	TIMESTAMP	-	NULL	Soft delete timestamp	NULL

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY (employee_id)
- UNIQUE KEY (username)
- UNIQUE KEY (email)
- INDEX (department_id)
- INDEX (zone_id)
- INDEX (is_active)

1.2 Table: roles

Purpose: Define system roles for RBAC

Field Name	Data Type	Size	Constraints	Description	Example
id	BIGINT	-	PK, AUTO, REQ	Unique role identifier	1
name	VARCHAR	100	UK, REQ	Role name (system identifier)	expenditure_control_officer
display_name	VARCHAR	100	REQ	Human-readable role name	Expenditure Control Officer
description	TEXT	-	NULL	Role description and responsibilities	Creates and manages payment vouchers
is_system_role	BOOLEAN	-	REQ, DEFAULT true	System-defined role (cannot be deleted)	true